Based on: Marco Avellaneda and Sasha Stoikov, High-frequency trading in a limit order book (2008).

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1 Introduction

1.1 Motivation for the Utility Function

A first idea would be to maximize the expected final wealth of the market maker. However, expected wealth alone does not account for risk.

Consider two market makers. Both expect to finish the day with a profit of 100.

- Market maker A has a 50% chance of making 500 and a 50% chance of losing 300.
- Market maker B makes many small trades and is almost certain to earn around 100.

Both have the same expected wealth:

$$0.5 \times 500 + 0.5 \times (-300) = 100.$$

If we only maximize expected wealth, these two strategies are considered equivalent. In practice, however, most market makers would prefer the second strategy because it achieves the same expected profit with much lower risk.

To incorporate this preference, instead of maximizing wealth directly, we maximize a utility function of wealth,

$$U(W).$$

The utility function measures the satisfaction associated with a given level of wealth.

We first require utility to increase with wealth:

$$U'(W) > 0.$$

More wealth should always be preferred to less wealth.

However, we do not want each additional euro to generate the same amount of satisfaction. Going from 0 to 100 is much more valuable than going from 1,000,000 to 1,000,100. Therefore, each extra euro should provide less satisfaction than the previous one.

Mathematically, this means that the utility function must be concave:

$$U''(W) < 0.$$

Equivalently, the tangent line lies above the graph of the function.

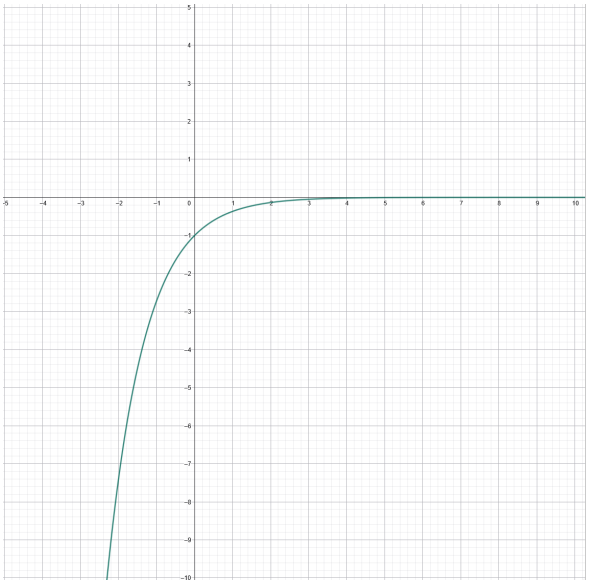


Figure 1: CARA utility function $U(W) = -e^{-\gamma W}$.

The ratio

$$-\frac{U''(W)}{U'(W)}$$

measures how quickly the marginal utility decreases relative to the utility itself. It is called the *absolute risk aversion*.

In the Avellaneda–Stoikov model, this quantity is assumed to be constant:

$$-\frac{U''(W)}{U'(W)} = \gamma, \qquad \gamma > 0.$$

This assumption is known as *Constant Absolute Risk Aversion* (CARA). A small value of γ corresponds to a less risk-averse market maker, while a large value of γ corresponds to a more risk-averse one.

Solving the differential equation above yields the exponential utility function

$$U(W) = -e^{-\gamma W},$$

which is the utility function used throughout the Avellaneda–Stoikov model.

1.2 Value Function Under Exponential Utility

Let X_t denote the cash held by the market maker and q_t the inventory. The marked-to-market wealth is

$$W_t = X_t + q_t S_t,$$

where S_t is the stock price.

Using the CARA utility introduced previously, the value function is defined as

$$u(x, s, q, t) = E_t\left[-e^{-\gamma W_T}\right] = E_t\left[-e^{-\gamma(X_T + q_T S_T)}\right].$$

To understand the role played by the exponential utility, consider a simplified setting where the inventory q remains constant and the stock price follows a Brownian motion

$$dS_t = \sigma dW_t.$$

Then

$$S_T = S_t + \sigma(W_T - W_t),$$

with

$$W_T - W_t \sim \mathcal{N}(0, T - t).$$

Conditionally on the information available at time t,

$$qS_T = qS_t + q\sigma\sqrt{T-t}\, Z,$$

where $Z \sim \mathcal{N}(0, 1)$.

The value function becomes

$$u(x, s, q, t) = -e^{-\gamma(x+qs)}\, E\left[e^{-\gamma q\sigma\sqrt{T-t}\, Z}\right].$$

Using the density of a standard normal random variable,

$$E[e^{aZ}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{az} e^{-z^2/2}\, dz.$$

Completing the square gives

$$E[e^{aZ}] = e^{a^2/2}.$$

With

$$a = -\gamma q\sigma\sqrt{T-t},$$

we obtain

$$E\left[e^{-\gamma q\sigma\sqrt{T-t}\,Z}\right] = \exp\left(\frac{\gamma^2 q^2\sigma^2(T-t)}{2}\right).$$

Hence

$$u(x,s,q,t) = -\exp\left(-\gamma(x+qs) + \frac{\gamma^2 q^2\sigma^2(T-t)}{2}\right).$$

The additional term

$$\frac{\gamma^2 q^2\sigma^2(T-t)}{2}$$

is the utility penalty associated with holding inventory. It increases with the risk aversion parameter γ, the inventory size q, the volatility σ, and the remaining time horizon $T-t$.

1.3 Reservation Bid and Ask Prices

The reservation prices are the prices at which the market maker is indifferent between executing a trade and keeping the current inventory.

The reservation bid price r^b is defined as the price at which the market maker is indifferent between:

- buying one additional share at price r^b,
- keeping the current inventory.

This indifference condition can be written as

$$u(x - r^b, s, q+1, t) = u(x,s,q,t).$$

Substituting the value function

$$u(x,s,q,t) = -\exp\left(-\gamma(x+qs) + \frac{\gamma^2 q^2\sigma^2(T-t)}{2}\right),$$

and solving for r^b yields

$$r^b(s,q,t) = s + \frac{(1-2q)\gamma\sigma^2(T-t)}{2}.$$

Similarly, the reservation ask price r^a is obtained from

$$u(x + r^a, s, q-1, t) = u(x,s,q,t),$$

which gives

$$r^a(s,q,t) = s + \frac{(-1-2q)\gamma\sigma^2(T-t)}{2}.$$

The reservation price is defined as the midpoint between the reservation bid and ask prices:

$$r(s,q,t) = \frac{r^a + r^b}{2} = s - q\gamma\sigma^2(T-t).$$

This quantity can be interpreted as the inventory-adjusted fair value of the asset.

If $q > 0$, the market maker is long inventory. In this case

$$r < s,$$

This reflects the desire to reduce inventory and lock in profits before adverse price movements occur.

As the terminal time T approaches,

$$T - t \to 0,$$

and therefore

$$r(s,q,t) \to s.$$

Near maturity there is little time left for inventory risk to materialize, so the inventory adjustment gradually disappears.

1.4 Infinite-Horizon Approximation

Instead of optimizing over a finite horizon, one may consider an infinite-horizon problem with discount rate $\omega > 0$.

The discounted value function is

$$\bar{u}(x,s,q) = E\left[\int_0^\infty e^{-\omega t}\left(-e^{-\gamma(X_t+qS_t)}\right)dt\right].$$

Using the finite-horizon expression derived previously,

$$u(x,s,q,t) = -\exp\left(-\gamma(x+qs) + \frac{\gamma^2 q^2\sigma^2 t}{2}\right),$$

we obtain

$$\bar{u}(x,s,q) = -e^{-\gamma(x+qs)}\int_0^\infty e^{-t\left(\omega - \frac{\gamma^2 q^2\sigma^2}{2}\right)}dt.$$

Provided

$$\omega > \frac{\gamma^2 q^2\sigma^2}{2},$$

the integral converges and

$$\bar{u}(x,s,q) = -\frac{e^{-\gamma(x+qs)}}{\omega - \frac{\gamma^2 q^2 \sigma^2}{2}}.$$

The corresponding infinite-horizon reservation prices are obtained from the same indifference conditions,

$$\bar{u}(x - r^b, s, q+1) = \bar{u}(x,s,q),$$

and

$$\bar{u}(x + r^a, s, q-1) = \bar{u}(x,s,q).$$

After simplification,

$$\bar{r}^a = s + \frac{1}{\gamma}\ln\left(1 + \frac{(1-2q)\gamma^2\sigma^2}{2\omega - \gamma^2 q^2 \sigma^2}\right),$$

and

$$\bar{r}^b = s + \frac{1}{\gamma}\ln\left(1 + \frac{(-1-2q)\gamma^2\sigma^2}{2\omega - \gamma^2 q^2 \sigma^2}\right).$$

Unlike the finite-horizon case, the inventory adjustment is now controlled by the discount factor ω, which determines the effective time scale over which inventory risk is taken into account.

2 Market Order Arrivals and Execution Intensities

We now consider an agent who can continuously post limit orders around the mid-price and seeks to maximize the expected utility of terminal wealth. The objective function is

$$u(s,x,q,t) = \max_{\delta_a,\delta_b} E_t\left[-e^{-\gamma(X_T + q_T S_T)}\right], \tag{1}$$

where S_t denotes the mid-price, X_t the cash process, q_t the inventory process, and $\gamma > 0$ the risk-aversion parameter.

To model the execution of limit orders, we assume that the size Q of incoming market orders follows a power-law distribution. Empirical studies suggest that market order sizes exhibit heavy tails and can be approximated by

$$f_Q(x) \propto x^{-1-\alpha}, \tag{2}$$

where the tail exponent α is typically observed to be around

$$\alpha \approx 1.5.$$

Let Λ denote the average arrival frequency of market orders. Following Avellaneda and Stoikov, we assume that the temporary market impact generated by a market order of size Q satisfies

$$\Delta p \approx \frac{1}{K} \ln(Q), \tag{3}$$

for some positive constant K.

A limit order posted at a distance δ from the mid-price is executed whenever the market impact exceeds that distance. Therefore, the execution intensity is

$$\lambda(\delta) = \Lambda \, P(\Delta p > \delta). \tag{4}$$

Using the logarithmic market impact assumption,

$$\Delta p > \delta \iff \frac{1}{K} \ln(Q) > \delta \tag{5}$$

$$\iff \ln(Q) > K\delta \tag{6}$$

$$\iff Q > e^{K\delta}. \tag{7}$$

Hence,

$$\lambda(\delta) = \Lambda \, P\left(Q > e^{K\delta}\right). \tag{8}$$

Using the power-law distribution of market order sizes,

$$P\left(Q > e^{K\delta}\right) = \int_{e^{K\delta}}^{\infty} x^{-1-\alpha} \, dx. \tag{9}$$

Evaluating the integral,

$$\int_{e^{K\delta}}^{\infty} x^{-1-\alpha} \, dx = \left[-\frac{1}{\alpha} x^{-\alpha} \right]_{e^{K\delta}}^{\infty} \tag{10}$$

$$= \frac{1}{\alpha} e^{-K\alpha\delta}. \tag{11}$$

Therefore,

$$\lambda(\delta) = \frac{\Lambda}{\alpha} e^{-K\alpha\delta}. \tag{12}$$

This expression shows that execution probabilities decrease exponentially as the quoted price moves away from the mid-price.

3 Derivation of the Hamilton–Jacobi–Bellman Equation

Recall that the market maker aims to maximize the expected utility of terminal wealth

$$u(s,x,q,t) = \max_{\delta_a,\delta_b} E_t\left[-e^{-\gamma(X_T+q_T S_T)}\right]. \tag{13}$$

Consider a small time interval dt. During this interval, three events may occur:

- a buy market order lifts our ask quote,
- a sell market order hits our bid quote,
- no trade occurs.

Since market orders arrive according to Poisson processes, the corresponding probabilities are

$$P(\text{ask fill}) = \lambda_a(\delta_a)dt,$$

$$P(\text{bid fill}) = \lambda_b(\delta_b)dt,$$

and

$$P(\text{no fill}) = 1 - (\lambda_a + \lambda_b)dt.$$

Using the definition of expectation as the sum of probabilities multiplied by the corresponding outcomes,

$$\begin{aligned} u(s,x,q,t) = {} & \lambda_a(\delta_a)dt\, u(s, x+s+\delta_a, q-1, t) \\ & + \lambda_b(\delta_b)dt\, u(s, x-s+\delta_b, q+1, t) \\ & + \Big(1 - (\lambda_a+\lambda_b)dt\Big) E_t[u_{\text{nofill}}]. \end{aligned} \tag{14}$$

To evaluate the last term, observe that

$$u_{t+dt} = u_t + du.$$

Therefore

$$E_t[u_{\text{nofill}}] = E_t[u_t + du] = u + E_t[du].$$

When no trade occurs, only the stock price changes according to

$$dS = \sigma dW.$$

Applying Itô's lemma,

$$du = u_t dt + u_s dS + \frac{1}{2} u_{ss} (dS)^2. \tag{15}$$

Substituting $dS = \sigma dW$,

$$du = u_t dt + \sigma u_s dW + \frac{1}{2} \sigma^2 u_{ss} dt.$$

Since

$$E_t[dW] = 0,$$

we obtain

$$E_t[du] = \left(u_t + \frac{1}{2} \sigma^2 u_{ss} \right) dt. \tag{16}$$

Hence

$$E_t[u_{\text{nofill}}] = u + \left(u_t + \frac{1}{2} \sigma^2 u_{ss} \right) dt. \tag{17}$$

Substituting into the expectation equation gives

$$\begin{aligned} u = {} & \lambda_a dt \, u_a + \lambda_b dt \, u_b \\ & + \Big(1 - (\lambda_a + \lambda_b) dt \Big) \left[u + \left(u_t + \frac{1}{2} \sigma^2 u_{ss} \right) dt \right], \end{aligned} \tag{18}$$

where

$$u_a = u(s, x + s + \delta_a, q - 1, t),$$

and

$$u_b = u(s, x - s + \delta_b, q + 1, t).$$

3.1 Exponential Utility Ansatz

We look for a solution of the form

$$u(s, x, q, t) = - \exp \left(-\gamma \big(x + \theta(s, q, t) \big) \right). \tag{19}$$

Differentiating with respect to time gives

$$u_t = -\gamma \theta_t u. \tag{20}$$

Similarly,

$$u_s = -\gamma \theta_s u, \tag{21}$$

and

$$u_{ss} = u\left(\gamma^2\theta_s^2 - \gamma\theta_{ss}\right). \tag{22}$$

Substituting into the HJB equation yields

$$-\gamma\theta_t u + \frac{1}{2}\sigma^2 u\left(\gamma^2\theta_s^2 - \gamma\theta_{ss}\right) + \lambda_b(u_b - u) + \lambda_a(u_a - u) = 0. \tag{23}$$

3.2 Reservation Bid Price

We define the reservation bid price as the amount of cash that makes the agent indifferent between holding q shares and $q+1$ shares,

$$u(s, x - r_b, q+1, t) = u(s, x, q, t). \tag{24}$$

Substituting the exponential ansatz,

$$-\exp\left(-\gamma\left(x - r_b + \theta(s, q+1, t)\right)\right) = -\exp\left(-\gamma\left(x + \theta(s, q, t)\right)\right). \tag{25}$$

Taking logarithms gives

$$r_b(s, q, t) = \theta(s, q+1, t) - \theta(s, q, t). \tag{26}$$

Likewise, the reservation ask price is defined by

$$u(s, x + r_a, q-1, t) = u(s, x, q, t), \tag{27}$$

which implies

$$r_a(s, q, t) = \theta(s, q, t) - \theta(s, q-1, t). \tag{28}$$

3.3 Execution Terms

Consider first the bid execution term

$$u_b = u(s, x - s + \delta_b, q+1, t). \tag{29}$$

Using the ansatz,

$$\begin{aligned} u_b &= -\exp\left(-\gamma\left(x - s + \delta_b + \theta(s, q+1, t)\right)\right) && (30)\\ &= u\exp\left(-\gamma\left(\delta_b + r_b - s\right)\right). && (31) \end{aligned}$$

Therefore

$$u_b - u = u\left(e^{-\gamma(\delta_b + r_b - s)} - 1\right). \tag{32}$$

Similarly,

$$u_a - u = u\left(e^{-\gamma(\delta_a + s - r_a)} - 1\right). \tag{33}$$

Substituting into the HJB and dividing by $-\gamma u$ yields

$$\theta_t + \frac{1}{2}\sigma^2\theta_{ss} - \frac{1}{2}\gamma\sigma^2\theta_s^2 + \frac{\lambda_b(\delta_b)}{\gamma}\left[1 - e^{-\gamma(\delta_b + r_b - s)}\right] \tag{34}$$

$$+ \frac{\lambda_a(\delta_a)}{\gamma}\left[1 - e^{-\gamma(\delta_a + s - r_a)}\right] = 0. \tag{35}$$

3.4 Optimal Bid Distance

We now optimize the bid contribution of the Hamilton–Jacobi–Bellman equation. Consider the function

$$f(\delta_b) = \frac{\lambda_b(\delta_b)}{\gamma}\left(1 - e^{-\gamma(\delta_b + r_b - s)}\right). \tag{36}$$

The optimal bid distance satisfies the first-order condition

$$\frac{\partial f}{\partial \delta_b} = 0. \tag{37}$$

Taking the derivative with respect to δ_b,

$$\frac{\partial f}{\partial \delta_b} = \frac{\lambda_b'(\delta_b)}{\gamma}\left(1 - e^{-\gamma(\delta_b + r_b - s)}\right)$$

$$+ \lambda_b(\delta_b)e^{-\gamma(\delta_b + r_b - s)} = 0. \tag{38}$$

Rearranging,

$$\frac{\lambda_b'(\delta_b)}{\gamma} = e^{-\gamma(\delta_b + r_b - s)}\left(\frac{\lambda_b'(\delta_b)}{\gamma} - \lambda_b(\delta_b)\right), \tag{39}$$

which gives

$$e^{-\gamma(\delta_b + r_b - s)} = \frac{\lambda_b'(\delta_b)}{\lambda_b'(\delta_b) - \gamma\lambda_b(\delta_b)}. \tag{40}$$

Dividing numerator and denominator by $\lambda_b'(\delta_b)$,

$$e^{-\gamma(\delta_b + r_b - s)} = \frac{1}{1 - \gamma\frac{\lambda_b(\delta_b)}{\lambda_b'(\delta_b)}}. \tag{41}$$

Taking the logarithm,

$$-\gamma(\delta_b + r_b - s) = -\ln\left(1 - \gamma\frac{\lambda_b(\delta_b)}{\lambda_b'(\delta_b)}\right). \tag{42}$$

Therefore, the optimality condition can be written as

$$\boxed{s - r_b = \delta_b - \frac{1}{\gamma}\ln\left(1 - \gamma\frac{\lambda_b(\delta_b)}{\lambda_b'(\delta_b)}\right)} \tag{43}$$

which corresponds to the implicit equation for the optimal bid distance.

3.5 Using the Exponential Arrival Intensity

Recall that the execution intensity is obtained from the power-law distribution of market order sizes:

$$\lambda_b(\delta_b) = \frac{\Lambda}{\alpha}e^{-K\alpha\delta_b}. \tag{44}$$

Therefore,

$$\lambda_b'(\delta_b) = -K\alpha\lambda_b(\delta_b). \tag{45}$$

Hence,

$$\frac{\lambda_b(\delta_b)}{\lambda_b'(\delta_b)} = -\frac{1}{K\alpha}. \tag{46}$$

Substituting into the optimality condition gives

$$s - r_b = \delta_b - \frac{1}{\gamma}\ln\left(1 + \frac{\gamma}{K\alpha}\right). \tag{47}$$

Thus,

$$\boxed{\delta_b^* = s - r_b + \frac{1}{\gamma}\ln\left(1 + \frac{\gamma}{K\alpha}\right)} \tag{48}$$

The same reasoning applies to the ask side, yielding

$$\boxed{r_a - s = \delta_a - \frac{1}{\gamma}\ln\left(1 - \gamma\frac{\lambda_a(\delta_a)}{\lambda_a'(\delta_a)}\right)} \tag{49}$$

and, using the same exponential intensity function,

$$\boxed{\delta_a^* = r_a - s + \frac{1}{\gamma}\ln\left(1 + \frac{\gamma}{K\alpha}\right)} \tag{50}$$

Substituting these optimal distances into the HJB equation gives the final reduced equation for $\theta(s,q,t)$:

$$\begin{aligned}\theta_t + \frac{1}{2}\sigma^2\theta_{ss} - \frac{1}{2}\gamma\sigma^2\theta_s^2 + \frac{\Lambda}{\alpha}e^{-K\alpha\delta_b^*}\left(1 - e^{-\gamma(\delta_b^* + r_b - s)}\right) \\ + \frac{\Lambda}{\alpha}e^{-K\alpha\delta_a^*}\left(1 - e^{-\gamma(\delta_a^* + s - r_a)}\right) \\ = \theta_t + \frac{1}{2}\sigma^2\theta_{ss} - \frac{1}{2}\gamma\sigma^2\theta_s^2 + \frac{\Lambda}{\alpha(K\alpha+\gamma)}\left(e^{-K\alpha\delta_b^*} + e^{-K\alpha\delta_a^*}\right) \\ = 0. \end{aligned} \tag{51}$$

4 Quadratic Approximation of the Value Function

To obtain a closed-form approximation, we assume that the function $\theta(s,q,t)$ can be approximated by a quadratic function of the inventory:

$$\theta(s,q,t) = \theta_0(s,t) + q\theta_1(s,t) + q^2\theta_2(s,t).$$

Using the definition of the reservation ask price,

$$\theta(s,q,t) - \theta(s,q-1,t),$$

we compute

$$\theta_0 + q\theta_1 + q^2\theta_2 - \left(\theta_0 + (q-1)\theta_1 + (q-1)^2\theta_2\right).$$

Therefore,

$$\theta(s,q,t) - \theta(s,q-1,t) = \theta_1 + (2q-1)\,\theta_2.$$

Similarly,

$$\theta(s,q+1,t) - \theta(s,q,t) = \theta_1 + (2q+1)\,\theta_2.$$

The reservation price, defined as the midpoint between the reservation bid and ask prices,

$$r(s,q,t) = \frac{r_a + r_b}{2},$$

is therefore

$$\boxed{r(s,q,t) = \theta_1(s,t) + 2q\theta_2(s,t)}$$

where θ_1 represents the reservation price when the inventory is zero, while θ_2 measures the sensitivity of the reservation price to inventory.

Since holding a positive inventory exposes the market maker to adverse price movements, θ_2 is typically negative.

4.1 Asymptotic approximation of $\theta(s, q, t)$

The terminal condition is given by the value of the inventory at maturity:

$$\theta(s, q, T) = qs$$

Therefore, by matching powers of q:

$$\theta_0(s, T) = 0, \qquad \theta_1(s, T) = s, \qquad \theta_2(s, T) = 0$$

4.1.1 First order term θ_1

The coefficient of q in the HJB equation gives:

$$\theta_{1,t} + \frac{1}{2}\sigma^2\theta_{1,ss} = 0$$

with terminal condition:

$$\theta_1(s, T) = s$$

Since:

$$\theta_{1,ss} = 0$$

we obtain:

$$\theta_{1,t} = 0$$

and therefore:

$$\boxed{\theta_1(s, t) = s}$$

4.1.2 Second order term θ_2

The coefficient of q^2 gives:

$$\theta_{2,t} + \frac{1}{2}\sigma^2\theta_{2,ss} - \frac{1}{2}\gamma\sigma^2 = 0$$

with:

$$\theta_2(s, T) = 0$$

Since θ_2 does not depend on s:

$$\theta_{2,ss} = 0$$

hence:

$$\theta_{2,t} = \frac{1}{2}\gamma\sigma^2$$

Integrating backward from maturity:

$$\theta_2(t) = \frac{1}{2}\gamma\sigma^2(t-T)$$

which gives:

$$\boxed{\theta_2(t) = -\frac{1}{2}\gamma\sigma^2(T-t)}$$

5 Reservation Price

The reservation price is defined as

$$r(s,q,t) = \frac{r_a + r_b}{2}.$$

Using the quadratic approximation,

$$r(s,q,t) = \theta_1 + 2q\theta_2,$$

and substituting the solutions,

$$\boxed{r(s,q,t) = s - q\gamma\sigma^2(T-t)}$$

which is the classical Avellaneda–Stoikov reservation price.

A positive inventory ($q > 0$) decreases the reservation price because the market maker wants to sell inventory and reduce exposure.

6 Optimal Spread

The spread is given by

$$r_a - r_b = -\theta_2 + \frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{K\alpha}\right).$$

Using

$$\theta_2 = -\gamma\sigma^2(T-t),$$

we obtain

$$\boxed{r_a - r_b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{K\alpha}\right)}$$

In the Avellaneda–Stoikov model, the exponential arrival intensity implies that the spread is independent of the inventory level q. Inventory only shifts the center of the quotes through the reservation price.